

4.12.22

AI25BTECH11017-BALU

Question:

The points $(3, 4)$ and $(2, -6)$ are situated on the _____ of the line $3x - 4y - 8 = 0$.

Solution:

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

Given two points

$$\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ -6 \end{pmatrix} \quad (0.1)$$

$$\mathbf{n} = \begin{pmatrix} \frac{-3}{4} \\ 1 \end{pmatrix} \quad (0.2)$$

Equation of line is

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.3)$$

$$\mathbf{n}^T \mathbf{a} - 1 = \frac{3}{4} \quad (0.4)$$

$$\mathbf{n}^T \mathbf{b} - 1 = \frac{-17}{2} \quad (0.5)$$

As the values are in opposite sign the points are situated in opposite direction w.r.t line

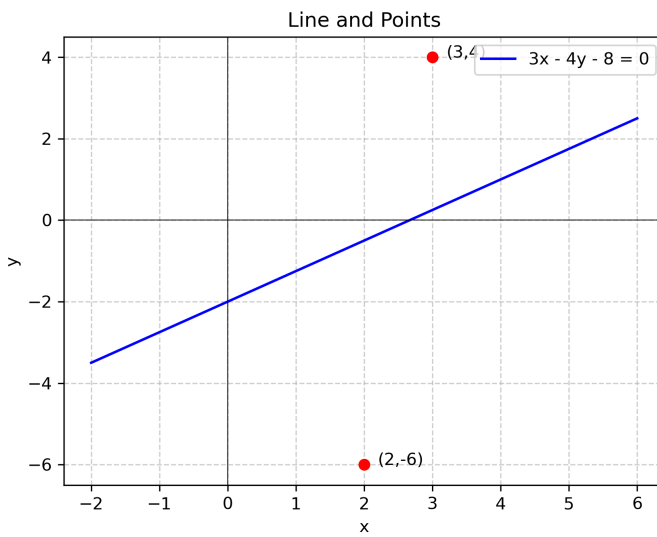


Fig. 0.1